Chapter 7. Complex Measures

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Overview

Results from Chapters 1–3 are cited as *Fact B.k* (see the *Background* notes). This chapter allows measures with negative or complex values. Relative to a positive measure, every measure splits into an integral part and a singular part (Lebesgue–Radon–Nikodym), the core of measure theory; applied to Lebesgue measure on the Borel sets, this describes all Borel measures on $\mathbb{R}$.

7.1 Signed Measures and Complex Measures

Motivation. Signed and complex measures are differences of positive measures (Hahn, Jordan), so the positive theory still applies.

7.1.1 Definitions

A function $\lambda : \mathcal{S} \to [0, \infty]$ on a σ-algebra $\mathcal{S}$ of a set X satisfying

$$\{A_n \in \mathcal{S} : n = 1, 2, \dots\} \text{ pairwise disjoint } \implies \lambda\Big(\bigcup_{n=1}^{\infty} A_n\Big) = \sum_{n=1}^{\infty} \lambda(A_n) \tag{1}$$

with $\lambda(\emptyset) = 0$ was called a positive measure.

Definition (signed measure, complex measure)

A function $\lambda : \mathcal{S} \to \mathbb{R}^*$ satisfying (1) and $\lambda(\emptyset) = 0$ is a **signed measure**. A function $\lambda : \mathcal{S} \to \mathbb{C}$ satisfying (1) is a **complex measure**.

Convergence of $\sum \lambda(A_n)$ is assumed (to $\pm\infty$ allowed for signed measures). If $\lambda(E) = \infty$ and $\lambda(F) = -\infty$, then in

$$\lambda(E \cup F) = \lambda(E \setminus F) + \lambda(F \setminus E) + \lambda(E \cap F)$$

both $\pm\infty$ would appear on the right: *a signed measure cannot take both values* $\pm\infty$. Complex measures form a vector space.

Example 1

For a measurable function h on $\mathbb{R}$ define

$$\lambda(E) = \int_E h(x)\, dx, \qquad E \in \mathcal{M}. \tag{2}$$

If $h \geq 0$, λ is a positive measure (Fact B.7); if $h : \mathbb{R} \to \mathbb{R}^*$ is measurable with h_+ or h_- integrable, (2) is a signed measure; if $h : \mathbb{R} \to \mathbb{C}$ is integrable, a complex measure.

Given a complex measure $\lambda : \mathcal{S} \to \mathbb{C}$, let $\lambda_1(E), \lambda_2(E)$ be the real and imaginary parts of $\lambda(E)$. Then λ_1, λ_2 are signed measures and

$$\lambda = \lambda_1 + i\lambda_2. \tag{3}$$

So we first study signed measures, showing that a signed measure is a difference of positive measures — just as a real function is $f = f_+ - f_-$.

7.1.2 The Hahn decomposition

Definition (positive and negative sets)

Let λ be a signed measure on a σ-algebra $\mathcal{S}$ of X. A set $P \in \mathcal{S}$ is **positive** if

$$E \in \mathcal{S},\ E \subset P \implies \lambda(E) \geq 0,$$

and **negative** if

$$E \in \mathcal{S},\ E \subset P \implies \lambda(E) \leq 0.$$

E.g. in Example 1, $\{h \geq 0\}$ and $\{h < 0\}$ are positive and negative sets. Measurable subsets of positive sets are positive, and a union P of positive sets P_n is positive: for $E \subset P$, $E \in \mathcal{S}$, put

$$E_n = (E \cap P_n) \setminus (P_1 \cup \cdots \cup P_{n-1}), \qquad n = 1, 2, \ldots \quad (P_0 = \emptyset);$$

then $E_n \subset P_n$, so $\lambda(E_n) \geq 0$, and $\lambda(E) = \sum_n \lambda(E_n) \geq 0$.

Theorem 7.1.1 (Hahn decomposition)

For a signed measure λ on $(X, \mathcal{S})$, X splits into a positive set and a negative set, and two such splittings (P_1, Q_1), (P_2, Q_2) satisfy, for every $E \in \mathcal{S}$,

$$\lambda(E \cap P_1) = \lambda(E \cap P_2), \qquad \lambda(E \cap Q_1) = \lambda(E \cap Q_2).$$

Proof. First assume λ does not take the value $+\infty$. Define

$$\alpha = \sup\{\lambda(P) : P \text{ is a positive set}\};$$

$\emptyset$ is positive, so $\alpha \geq 0$. Choose positive sets $\langle P_n \rangle$ with $\alpha = \lim_n \lambda(P_n)$; their union P is positive. For each n, $P \setminus P_n \subset P$ is positive, so $\lambda(P \setminus P_n) \geq 0$ and

$$\lambda(P_n) \leq \lambda(P_n) + \lambda(P \setminus P_n) = \lambda(P) \leq \alpha.$$

Hence $\lambda(P) = \alpha < \infty$. Now put $Q = X \setminus P$. If a measurable $E \subset Q$ is positive, then $E \cup P$ is positive, so

$$\alpha \geq \lambda(E \cup P) = \lambda(E) + \lambda(P) = \lambda(E) + \alpha \geq \alpha,$$

and therefore $\lambda(E) = 0$. That is,

$$E \in \mathcal{S},\ E \subset Q,\ E \text{ positive} \implies \lambda(E) = 0. \tag{4}$$

It remains to show Q is negative. Let $E \subset Q$, $E \in \mathcal{S}$. If E is positive, $\lambda(E) = 0$ by (4). Otherwise let n_1 be the least natural number with some $F_1 \subset E$, $\lambda(F_1) < -\frac{1}{n_1}$. If $E \setminus F_1$ is positive, $\lambda(E \setminus F_1) = 0$ by (4) and $\lambda(E) = \lambda(F_1) < 0$. Otherwise repeat: if $E \setminus (F_1 \cup \cdots \cup F_{k-1})$ is positive, $\lambda(E) = \sum_{i<k} \lambda(F_i) < 0$; else take the least n_k and $F_k \in \mathcal{S}$ with

$$F_k \subset E \setminus (F_1 \cup \cdots \cup F_{k-1}), \qquad \lambda(F_k) < -\frac{1}{n_k}.$$

If the process never stops, put

$$F = E \setminus \Big[\bigsqcup_{k=1}^{\infty} F_k\Big].$$

Clearly $\langle n_k \rangle$ is nondecreasing.

If $\langle n_k \rangle$ is bounded, then $\lambda(\bigsqcup_k F_k) = -\infty$, and since $\lambda(F) < \infty$, $\lambda(E) = \lambda(\bigsqcup_k F_k) + \lambda(F) = -\infty$, and we are done.

If $\langle n_k \rangle$ is unbounded, we show F is positive. Suppose F contains $G \in \mathcal{S}$ with $\lambda(G) < 0$. Since $n_k \to \infty$, we can choose k with $\frac{1}{n_k - 1} < -\lambda(G)$. Then

$$G \subset F \subset E \setminus (F_1 \cup \cdots \cup F_{k-1}), \qquad \lambda(G) < -\frac{1}{n_k - 1},$$

contradicting the minimality in the choice of n_k. So F is positive, $\lambda(F) = 0$ by (4), and $\lambda(E) = \sum_{k=1}^{\infty} \lambda(F_k) < 0$. Hence Q is negative.

The case where λ does not take $-\infty$ is proved in the same way (apply the above to $-\lambda$).

For the second part, see Exercise 7.1.1. ■

Exercise 7.1.1

Prove the second part of the Hahn decomposition theorem.

Solution [Supplement]**.** For $E \in \mathcal{S}$, $E \cap (P_1 \setminus P_2) = E \cap P_1 \cap Q_2$ lies in the positive P_1 and the negative Q_2, so has measure 0; likewise $E \cap (P_2 \setminus P_1)$. Hence

$$\lambda(E \cap P_1) = \lambda(E \cap P_1 \cap P_2) + \lambda(E \cap P_1 \cap Q_2) = \lambda(E \cap P_1 \cap P_2) = \lambda(E \cap P_2).$$

The same argument with Q_1, Q_2 gives $\lambda(E \cap Q_1) = \lambda(E \cap Q_2)$. □

7.1.3 Jordan decomposition and total variation

Given a Hahn decomposition $X = P \sqcup Q$ for λ, define for $E \in \mathcal{S}$

$$\lambda_+(E) = \lambda(E \cap P), \qquad \lambda_-(E) = -\lambda(E \cap Q).$$

By Theorem 7.1.1 these do not depend on the decomposition. λ_+ and λ_- are positive measures and

$$\lambda(E) = \lambda_+(E) - \lambda_-(E), \qquad E \in \mathcal{S}.$$

Also define

$$|\lambda|(E) = \lambda_+(E) + \lambda_-(E);$$

$|\lambda|$ is a positive measure as well.

Exercise 7.1.2

If λ is defined by (2), show that for every $E \in \mathcal{M}$

$$\lambda_+(E) = \int_E h_+, \qquad \lambda_-(E) = \int_E h_-, \qquad |\lambda|(E) = \int_E |h|.$$

Solution [Supplement]**.** $P = \{h \geq 0\}$, $Q = \{h < 0\}$ is a Hahn decomposition: for $E \subset P$, $\lambda(E) = \int_E h \geq 0$, and for $E \subset Q$, $\lambda(E) \leq 0$. Since $h\chi_P = h_+$ and $h\chi_Q = -h_-$,

$$\lambda_+(E) = \int_{E \cap P} h = \int_E h_+, \qquad \lambda_-(E) = -\int_{E \cap Q} h = \int_E h_-,$$

and $|\lambda|(E) = \int_E (h_+ + h_-) = \int_E |h|$. □

Proposition 7.1.2

If λ is a signed measure on a σ-algebra $\mathcal{S}$ of X, then for every $E \in \mathcal{S}$

$$|\lambda|(E) = \sup\Big\{\sum_{n=1}^{\infty}|\lambda(E_n)| : \ E_n \in \mathcal{S},\ E = \bigsqcup_{n=1}^{\infty} E_n\Big\}. \tag{5}$$

Proof. Let α be the right side of (5). If (P, Q) is a Hahn decomposition, then

$$|\lambda|(E) = \lambda_+(E) + \lambda_-(E) = |\lambda(E \cap P)| + |\lambda(E \cap Q)| \le \alpha.$$

Conversely, if $E_n \in \mathcal{S}$ and $E = \bigsqcup_{n=1}^{\infty} E_n$, then

$$\sum_{n=1}^{\infty}|\lambda(E_n)| = \sum_{n=1}^{\infty}|\lambda_+(E_n) - \lambda_-(E_n)| \le \sum_{n=1}^{\infty}(\lambda_+(E_n) + \lambda_-(E_n)) = \sum_{n=1}^{\infty}|\lambda|(E_n) = |\lambda|(E),$$

so $\alpha \le |\lambda|(E)$. ∎

7.1.4 Total variation of a complex measure

From now on let λ be a complex measure on a σ-algebra $(X, \mathcal{S})$ and define $|\lambda|(E)$ for $E \in \mathcal{S}$ by (5). Clearly $|\lambda(E)| \le |\lambda|(E)$ for every $E \in \mathcal{S}$.

Proposition 7.1.3

If λ is a complex measure on a σ-algebra $\mathcal{S}$ of X, then $E \mapsto |\lambda|(E)$ is a positive measure.

Proof. For a disjoint sequence $\langle E_n\rangle$ in $\mathcal{S}$ with union E we must show

$$|\lambda|(E) = \sum_{n=1}^{\infty}|\lambda|(E_n).$$

If $E = \bigsqcup_{m=1}^{\infty} F_m$, then for all m, n

$$F_m = \bigsqcup_{n=1}^{\infty}(E_n \cap F_m), \qquad E_n = \bigsqcup_{m=1}^{\infty}(E_n \cap F_m).$$

Hence

$$\begin{aligned}\sum_{m=1}^{\infty}|\lambda(F_m)| &= \sum_{m=1}^{\infty}\Big|\sum_{n=1}^{\infty}\lambda(E_n \cap F_m)\Big| \le \sum_{m=1}^{\infty}\sum_{n=1}^{\infty}|\lambda(E_n \cap F_m)| \\ &= \sum_{n=1}^{\infty}\sum_{m=1}^{\infty}|\lambda(E_n \cap F_m)| \le \sum_{n=1}^{\infty}|\lambda|(E_n),\end{aligned}$$

and taking the supremum over partitions $\{F_m\}$ gives $|\lambda|(E) \le \sum_n |\lambda|(E_n)$. For the reverse, let $\epsilon > 0$ and choose for each E_n a partition $\{E_{n,m} : m \in \mathbb{N}\}$ with

$$|\lambda|(E_n) < \sum_{m=1}^{\infty}|\lambda(E_{n,m})| + \frac{\epsilon}{2^n}, \qquad n = 1, 2, \dots$$

Then $\{E_{n,m} : m, n = 1, 2, \dots\}$ is a partition of E, so

$$\sum_{n=1}^{\infty}|\lambda|(E_n) \le \sum_{m,n=1}^{\infty}|\lambda(E_{n,m})| + \epsilon \le |\lambda|(E) + \epsilon.$$

■

By (3), a complex measure λ is a linear combination of positive measures:

$$\lambda = (\lambda_{1+} - \lambda_{1-}) + i(\lambda_{2+} - \lambda_{2-}). \tag{6}$$

If $\{E_n\}$ is a partition of $E \in \mathcal{S}$, then

$$\sum_{n=1}^{\infty} |\lambda(E_n)| \le \sum_{n=1}^{\infty} \left(\lambda_{1+}(E_n)+\lambda_{1-}(E_n)+\lambda_{2+}(E_n)+\lambda_{2-}(E_n)\right) = \lambda_{1+}(E)+\lambda_{1-}(E)+\lambda_{2+}(E)+\lambda_{2-}(E),$$

so for every $E \in \mathcal{S}$

$$|\lambda|(E) \le \lambda_{1+}(E) + \lambda_{1-}(E) + \lambda_{2+}(E) + \lambda_{2-}(E).$$

Since λ is a complex measure, the values on the right are always finite. Hence $|\lambda|(E) < \infty$ for every $E \in \mathcal{S}$; in particular $|\lambda|(X) < \infty$, and we define

$$\|\lambda\| = |\lambda|(X).$$

Positive measures are in general not complex measures; **finite positive measures** ($\mu(X) < \infty$) are.

Exercise 7.1.3

For a σ-algebra $\mathcal{S}$ on X, show that the set of all complex measures on $\mathcal{S}$ is a vector space and that $\lambda \mapsto \|\lambda\|$ is a norm on it.

Solution [Supplement]**.** If λ, ρ are complex measures and $a \in \mathbb{C}$, then $(\lambda+\rho)(\emptyset) = 0$ and for disjoint A_n,

$$(\lambda+\rho)\Big(\bigsqcup A_n\Big) = \sum \lambda(A_n) + \sum \rho(A_n) = \sum (\lambda+\rho)(A_n)$$

(sum of convergent series); similarly for $a\lambda$. So they form a vector space. Norm: $\|\lambda\| \ge 0$. If $\|\lambda\| = 0$ then $|\lambda(E)| \le |\lambda|(E) \le |\lambda|(X) = 0$ for all E, so $\lambda = 0$. For any partition $\{E_n\}$ of X, $\sum |a\lambda(E_n)| = |a| \sum |\lambda(E_n)|$, so $|a\lambda| = |a||\lambda|$ and $\|a\lambda\| = |a|\|\lambda\|$. Finally $\sum |(\lambda+\rho)(E_n)| \le \sum |\lambda(E_n)| + \sum |\rho(E_n)| \le \|\lambda\| + \|\rho\|$; taking the supremum gives $\|\lambda+\rho\| \le \|\lambda\| + \|\rho\|$. □

7.2 The Radon–Nikodym Theorem and the Lebesgue Decomposition

Motivation. $\nu(E) = \int_E f\, d\mu$ (a density) holds exactly when ν ignores μ-null sets; this underlies densities, conditional expectation and the dual of L^p.

7.2.1 Absolute continuity and singularity

Definition

Let μ be a positive measure and λ a signed or complex measure on a σ-algebra $\mathcal{S}$ of X.

- λ is **absolutely continuous** with respect to μ, written $\lambda \ll \mu$, if

$$E \in \mathcal{S},\ \mu(E) = 0 \implies \lambda(E) = 0.$$

- λ is **singular** with respect to μ, written $\lambda \perp \mu$, if there are disjoint $A, B \in \mathcal{S}$ with

$$\lambda(E) = \lambda(A \cap E), \qquad \mu(E) = \mu(B \cap E), \qquad E \in \mathcal{S}. \tag{7}$$

(2) is absolutely continuous with respect to Lebesgue measure; the Dirac measure is singular.

Linear combinations of measures $\ll \mu$ are $\ll \mu$. For $\lambda_1, \lambda_2 \perp \mu$ take disjoint $A_i, B_i \in \mathcal{S}$ with, for $i = 1, 2$,

$$\lambda_i(E) = \lambda_i(A_i \cap E), \qquad \mu(E) = \mu(B_i \cap E), \qquad E \in \mathcal{S}.$$

Put $A = A_1 \cup A_2$, $B = B_1 \cap B_2$. Then A, B are disjoint and

$$(\lambda_1 + \lambda_2)(A \cap E) = (\lambda_1 + \lambda_2)(E), \qquad \mu(B \cap E) = \mu(E)$$

(indeed $\lambda_1(A \cap E) = \lambda_1(A_1 \cap A \cap E) = \lambda_1(A_1 \cap E)$, and $\mu(E) = \mu(B_1 \cap E) = \mu(B_2 \cap B_1 \cap E)$). So linear combinations of singular measures are singular.

If a complex measure λ is both absolutely continuous and singular with respect to μ, take disjoint A, B satisfying (7). For each $E \in \mathcal{S}$, $\mu(E \cap A) = \mu(E \cap A \cap B) = 0$, so $\lambda(E \cap A) = 0$ by $\lambda \ll \mu$. Hence $\lambda(E) = \lambda(E \cap A) = 0$, i.e. $\lambda = 0$.

Fix a positive measure μ. Part (8) of the next theorem, the **Lebesgue decomposition**, splits every complex measure into absolutely continuous and singular parts; part (9) gives the absolutely continuous part by a function, the **Radon–Nikodym derivative**. This theorem makes precise "measures are generalized functions".

Theorem 7.2.1 (Lebesgue–Radon–Nikodym)

Let μ be a σ-finite positive measure and λ a complex measure on a σ-algebra $\mathcal{S}$ of X. Then there are unique measures λ_a, λ_s with

$$\lambda = \lambda_a + \lambda_s, \qquad \lambda_a \ll \mu, \qquad \lambda_s \perp \mu. \tag{8}$$

Moreover there is a unique $h \in L^1(X, \mu)$ with

$$\lambda_a(E) = \int_E h \, d\mu, \qquad E \in \mathcal{S}. \tag{9}$$

If μ and λ are σ-finite positive measures, there are unique measures λ_a, λ_s satisfying (8) and a unique measurable $h : X \to [0, \infty]$ satisfying (9).

Proof. Let λ be complex; first uniqueness. Two decompositions give $\lambda_a - \lambda_a' = \lambda_s' - \lambda_s$, both $\ll \mu$ and $\perp \mu$, hence 0; $h \in L^1(\mu)$ is unique by Fact B.8(b).

Step 1: μ and λ finite positive measures. Then $\nu = \mu + \lambda$ is also a finite positive measure. For every $f \in L^2(\nu)$, by the Cauchy–Schwarz inequality,

$$\Big| \int_X f \, d\lambda \Big| \le \int_X |f| \, d\lambda \le \int_X |f| \, d\nu \le \Big(\int_X |f|^2 \, d\nu \Big)^{1/2} \nu(X)^{1/2},$$

and since $\nu(X) < \infty$, $f \mapsto \int_X f \, d\lambda$ is a bounded linear functional on $L^2(\nu)$. By Theorem 6.3.3 (Riesz) there is $g \in L^2(\nu)$ with

$$\int_X f \, d\lambda = \int_X f g \, d\nu, \qquad f \in L^2(\nu). \tag{10}$$

As $\nu(X) < \infty$, Fact B.29 gives $g \in L^1(\nu)$. Moreover, putting $f = \chi_E$,

$$\nu(E) > 0 \implies \frac{1}{\nu(E)} \int_E g \, d\nu = \frac{\lambda(E)}{\nu(E)} \in [0, 1],$$

so by Fact B.27, $0 \le g \le 1$ ν-almost everywhere. Hence we may assume $0 \le g \le 1$ while keeping (10). Put

$$A = \{x \in X : 0 \le g(x) < 1\}, \qquad B = \{x \in X : g(x) = 1\},$$

and for $E \in \mathcal{S}$

$$\lambda_a(E) = \lambda(E \cap A), \qquad \lambda_s(E) = \lambda(E \cap B).$$

Rewriting (10) (using $\nu = \mu + \lambda$),

$$\int_X (1-g) f \, d\lambda = \int_X fg \, d\mu, \qquad f \in L^2(\nu). \tag{11}$$

Putting $f = \chi_B$,

$$0 = \int_B (1-g)\, d\lambda = \int_B g \, d\mu = \mu(B),$$

so $\lambda_s \perp \mu$ (with the disjoint sets B and $A = X \setminus B$: $\lambda_s(E) = \lambda_s(B \cap E)$ and $\mu(E) = \mu(A \cap E)$). Next, putting $f = (1 + g + \cdots + g^n)\chi_E$ in (11),

$$\int_E (1 - g^{n+1})\, d\lambda = \int_E g(1 + g + \cdots + g^n)\, d\mu.$$

Let $n \to \infty$: as $(1 - g^{n+1})|_B = 0$ and $1 - g^{n+1} \nearrow 1$ on A, the left side tends to $\lambda(E \cap A) = \lambda_a(E)$, and with $h = \lim_n g(1 + g + \cdots + g^n)$ the right side to $\int_E h \, d\mu$ (monotone convergence). This gives (8), (9), $\lambda_a \ll \mu$, and $h \in L^1(\mu)$ as $\lambda_a(X) < \infty$.

Step 2: μ σ-finite, λ finite positive. Write $X = \bigsqcup_n X_n$ with $\mu(X_n) < \infty$ and choose positive numbers $\langle \alpha_n \rangle$ with $\sum_n \alpha_n \mu(X_n) < \infty$. For $E \in \mathcal{S}$ put

$$\mu_0(E) = \sum_{n=1}^{\infty} \alpha_n \mu(E \cap X_n); \tag{12}$$

μ_0 is clearly a finite positive measure. By Step 1 there are λ_a, λ_s with

$$\lambda = \lambda_a + \lambda_s, \qquad \lambda_a \ll \mu_0, \qquad \lambda_s \perp \mu_0.$$

If $\mu(E) = 0$ then $\mu_0(E) = 0$, so $\lambda_a \ll \mu_0$ gives $\lambda_a \ll \mu$. If $\mu_0(E \setminus B) = 0$, then $\mu((E \setminus B) \cap X_n) = 0$ for all n ($\alpha_n > 0$), so $\mu(E \setminus B) = 0$; thus $\lambda_s \perp \mu_0$ gives $\lambda_s \perp \mu$, proving (8). By Step 1 there is $h_0 \in L^1(\mu_0)$ with

$$\lambda_a(E) = \int_E h_0 \, d\mu_0, \qquad E \in \mathcal{S}.$$

Put

$$k = \sum_{n=1}^{\infty} \alpha_n \chi_{X_n}. \tag{13}$$

Then $\int_E d\mu_0 = \int_E k \, d\mu$, so

$$\int_X f \, d\mu_0 = \int_X fk \, d\mu$$

for simple f, hence for measurable $f \geq 0$. With $f = h_0 \chi_E$ and $h = h_0 k$ this gives (9), and $\int h \, d\mu = \int h_0 \, d\mu_0 < \infty$ ($h_0 \geq 0$).

This proves the theorem for σ-finite positive μ and finite positive λ; complex λ are combinations of four such by (6) (uniqueness as above). ∎

Exercise 7.2.1

Prove the Lebesgue–Radon–Nikodym theorem for σ-finite positive λ, and for signed λ that are differences of σ-finite positive measures. (Uniqueness cannot subtract possibly infinite measures.)

Solution [Supplement]. *Existence (λ σ-finite positive).* Write $X = \bigsqcup_m Y_m$, $\lambda(Y_m) < \infty$, and $\lambda^{(m)}(E) = \lambda(E \cap Y_m)$. The theorem gives $\lambda^{(m)} = \lambda_a^{(m)} + \lambda_s^{(m)}$, $d\lambda_a^{(m)} = h_m\,d\mu$, $h_m \geq 0$, $\lambda_s^{(m)}$ concentrated on A_m, $\mu(A_m) = 0$; we may take $A_m \subset Y_m$ and $h_m = h_m \chi_{Y_m}$. Put

$$\lambda_a = \sum_m \lambda_a^{(m)}, \quad \lambda_s = \sum_m \lambda_s^{(m)}, \quad h = \sum_m h_m, \quad A = \bigcup_m A_m.$$

Then $\lambda = \lambda_a + \lambda_s$, $\lambda_a(E) = \int_E h\,d\mu$ (monotone convergence), $\lambda_a \ll \mu$, $\mu(A) = 0$ and $\lambda_s(E) = \lambda_s(E \cap A)$, so $\lambda_s \perp \mu$.

Uniqueness. For another decomposition with h', A', $\mu(A') = 0$, let $N = A \cup A'$ ($\mu(N) = 0$). Singular parts vanish on N^c and absolutely continuous parts on N, so for every E

$$\lambda_s(E) = \lambda(E \cap N) = \lambda_s'(E), \qquad \lambda_a(E) = \lambda(E \setminus N) = \lambda_a'(E).$$

No subtraction was used. For h: λ_a is finite on each $Y_m \cap X_n$ ($\mu(X_n) < \infty$), so $\int_E (h - h')\,d\mu = 0$ for $E \subset Y_m \cap X_n$, and $h = h'$ a.e.

Signed case. For $\lambda = \rho_1 - \rho_2$ with σ-finite positive ρ_i (one finite), decompose each ρ_i and set $\lambda_a = \rho_{1,a} - \rho_{2,a}$, $\lambda_s = \rho_{1,s} - \rho_{2,s}$, $h = h_1 - h_2$; these are well defined, $\ll \mu$ and $\perp \mu$ respectively. Uniqueness: $\lambda_s(E) = \lambda(E \cap N)$ and $\lambda_a(E) = \lambda(E \setminus N)$ for *any* decomposition. □

When λ_a and μ satisfy (9) we write

$$d\lambda_a = h\,d\mu, \qquad h = \Big[\frac{d\lambda_a}{d\mu}\Big].$$

Exercise 7.2.2

If $d\lambda = g\,d\mu$, show that $\lambda(\{x : g(x) = 0\}) = 0$.

Solution [Supplement]. Let $Z = \{x : g(x) = 0\}$. Then $\lambda(Z) = \int_Z g\,d\mu = \int_Z 0\,d\mu = 0$. (More generally $\lambda(E) = 0$ and even $|\lambda|(E) = 0$ for every measurable $E \subset Z$.) □

7.2.2 Polar decomposition

The Lebesgue–Radon–Nikodym theorem has many applications in measure theory; we look at a few.

Theorem 7.2.2 (polar decomposition)

For every complex measure λ on a set X there is a unique $h \in L^1(X, |\lambda|)$ with

$$d\lambda = h\,d|\lambda|, \qquad |h| = 1.$$

Proof. Since $\lambda \ll |\lambda|$, there is a unique $h \in L^1(|\lambda|)$ with $d\lambda = h\,d|\lambda|$ (Theorem 7.2.1, the singular part being 0). If $|\lambda|(E) > 0$, then

$$\Big|\frac{1}{|\lambda|(E)} \int_E h\,d|\lambda|\Big| = \frac{|\lambda(E)|}{|\lambda|(E)} \leq 1,$$

so by Fact B.27, $|h| \leq 1$ almost everywhere. On the other hand, for $r > 0$ put $A_r = \{x \in X : |h(x)| < r\}$. If $A_r = \bigsqcup_{n=1}^{\infty} E_n$, then

$$\sum_{n=1}^{\infty} |\lambda(E_n)| = \sum_{n=1}^{\infty} \Big|\int_{E_n} h\,d|\lambda|\Big| \leq \sum_{n=1}^{\infty} r|\lambda|(E_n) = r|\lambda|(A_r),$$

so $|\lambda|(A_r) \leq r|\lambda|(A_r)$, and for $r < 1$ this forces $|\lambda|(A_r) = 0$. Hence $|h| \geq 1$ almost everywhere, and we may redefine h on a null set so that $|h| = 1$. ■

7.2.3 Integration with respect to a complex measure

The polar decomposition suggests how to define integration with respect to a complex measure. Given $d\lambda = h\, d|\lambda|$, for every $f \in L^1(X, |\lambda|)$ define

$$\int_X f\, d\lambda = \int_X fh\, d|\lambda|.$$

If we decompose λ into positive measures as in (6), then for every $f \in L^1(X, |\lambda|)$

$$\int_X f\, d\lambda = \Big(\int_X f\, d\lambda_{1+} - \int_X f\, d\lambda_{1-}\Big) + i\Big(\int_X f\, d\lambda_{2+} - \int_X f\, d\lambda_{2-}\Big). \tag{14}$$

For characteristic functions this is the definition of $d\lambda = h\, d|\lambda|$, hence it holds for simple functions. For $0 \le f \in L^1(X, |\lambda|)$ take simple $s_n \nearrow f$ (Fact B.5); as $f \in L^1(|\lambda|)$ and $f \in L^1(\lambda_{i\pm})$ ($\lambda_{i\pm} \le |\lambda|$), dominated convergence gives (14), hence (14) for all complex $f \in L^1(X, |\lambda|)$.

7.2.4 The dual of $L^p(\mu)$

Finally, another application of Radon–Nikodym: the dual of $L^p(\mu)$ is $L^q(\mu)$ ($1 \le p < \infty$); again σ-finiteness is needed. Recall from Example 2 of §6.1 that

$$g \mapsto \varphi_g : L^q(\mu) \to L^p(\mu)^*$$

is an isometry (using σ-finiteness for $p = 1$, $q = \infty$); we show it is onto: for $\varphi \in L^p(\mu)^*$ we find $g \in L^q(\mu)$ with $\varphi = \varphi_g$.

Finite μ. For $E \in \mathcal{S}$ put

$$\lambda(E) = \varphi(\chi_E).$$

Since $\chi_{E_1 \sqcup E_2} = \chi_{E_1} + \chi_{E_2}$, $\lambda(E_1 \sqcup E_2) = \lambda(E_1) + \lambda(E_2)$. If $E = \bigsqcup_{n=1}^{\infty} E_n$ and $A_n = E_1 \sqcup \cdots \sqcup E_n$, then

$$\lim_{n\to\infty} \|\chi_E - \chi_{A_n}\|_p = \lim_{n\to\infty} \left[\mu(E \setminus A_n)\right]^{1/p} = 0$$

(here $p < \infty$ and $\mu(E) < \infty$), and since $\varphi \in L^p(\mu)^*$,

$$\lambda(E) = \lim_{n\to\infty} \lambda(A_n) = \lim_{n\to\infty} \sum_{k=1}^{n} \lambda(E_k).$$

So λ is a complex measure. Clearly $\lambda \ll \mu$ (if $\mu(E) = 0$ then $\chi_E = 0$ in L^p), so taking the Radon–Nikodym derivative $g \in L^1(\mu)$,

$$\varphi(\chi_E) = \int_E g\, d\mu = \int_X \chi_E g\, d\mu = \varphi_g(\chi_E), \qquad E \in \mathcal{S}.$$

So $\varphi = \varphi_g$ on simple functions, dense in $L^\infty(\mu)$, where both are continuous ($\|f\|_p \le \mu(X)^{1/p}\|f\|_\infty$, $g \in L^1$); so $\varphi = \varphi_g$ on $L^\infty(\mu)$. It remains to show $g \in L^q(\mu)$ and $\varphi = \varphi_g$ on $L^p(\mu)$.

Exercise 7.2.3

For a finite positive measure μ, show that the vector space of simple functions is dense in $L^p(\mu)$ for $1 \le p \le \infty$. [Hint: treat $1 \le p < \infty$ and $p = \infty$ separately; Fact B.20.]

Solution [Supplement]. $p = \infty$. Take $|f| \le \|f\|_\infty = M$ everywhere. Cover $\{|z| \le M\}$ by disjoint Borel $B_1, \dots, B_N$ of diameter $< \epsilon$, pick $z_k \in B_k$, and let $s = \sum_k z_k \chi_{f^{-1}(B_k)}$; then $\|f - s\|_\infty \le \epsilon$ (no finiteness of μ needed). $1 \le p < \infty$. Take $f \ge 0$ and simple $0 \le s_n \nearrow f$ (Fact B.5). $|f - s_n|^p \le f^p \in L^1$ tends to 0, so $\|f - s_n\|_p \to 0$ by dominated convergence. (Finiteness of μ puts all simple functions, e.g. χ_X, in L^p.) □

$p = 1$, $q = \infty$. For every $E \in \mathcal{S}$,

$$\Big|\int_E g\,d\mu\Big| = |\varphi(\chi_E)| \le \|\varphi\|\|\chi_E\|_1 = \|\varphi\|\mu(E),$$

so by Fact B.27, $\|g\|_\infty \le \|\varphi\|$.

$1 < p < \infty$. The basic method is the same as for ℓ^p. For $n = 1, 2, \dots$ put $E_n = \{x \in X : |g(x)| \le n\}$ and $f_n = \chi_{E_n}|g|^{q-1}\cdot\operatorname{sgn} g$ (with $\operatorname{sgn} g = \overline{g}/|g|$ where $g \neq 0$, so that $f_n g = \chi_{E_n}|g|^q$). Then $f_n \in L^\infty(\mu)$ and

$$\int_{E_n}|g|^q = \int_X f_n g = \varphi_g(f_n) = \varphi(f_n) \le \|\varphi\|\|f_n\|_p = \|\varphi\|\Big(\int_{E_n}|g|^q\Big)^{1/p}.$$

Dividing both sides by the last factor and applying the monotone convergence theorem,

$$\|g\|_q = \lim_{n\to\infty}\Big(\int_X \chi_{E_n}|g|^q\Big)^{1/q} \le \|\varphi\|.$$

Now that $g \in L^q(\mu)$, φ_g is continuous on $L^p(\mu)$; since simple functions are dense in $L^p(\mu)$, $\varphi(f) = \varphi_g(f)$ for every $f \in L^p(\mu)$. So $L^p(\mu)^* = L^q(\mu)$ for finite μ and $1 \le p < \infty$.

σ-finite μ. As in the Lebesgue–Radon–Nikodym theorem, take μ_0 and $k \in L^1(\mu)$ from (12), (13), $d\mu_0 = k\,d\mu$. Then

$$K : f \mapsto k^{1/p} f : L^p(\mu_0) \to L^p(\mu)$$

is a surjective isometry, so $\varphi \circ K$ is a bounded linear functional on $L^p(\mu_0)$. By the finite case there is $g \in L^q(\mu_0)$ with

$$(\varphi \circ K)(f) = \int_X f g\,d\mu_0, \qquad f \in L^p(\mu_0).$$

Rewriting,

$$\varphi(Kf) = \int_X fgk\,d\mu = \int_X k^{1/p} f g k^{1/q}\,d\mu = \int_X (Kf)\cdot(gk^{1/q})\,d\mu$$

(with $1/\infty = 0$). Hence $gk^{1/q}$ is the function we were looking for.

Exercise 7.2.4

In the argument above, show that K is a surjective isometry from $L^p(\mu_0)$ onto $L^p(\mu)$, and that $gk^{1/q} \in L^q(\mu)$.

Solution [Supplement]. $k > 0$ everywhere ($\alpha_n > 0$). *Isometry:* $\int_X |k^{1/p} f|^p\,d\mu = \int_X |f|^p\,d\mu_0$. *Surjective:* for $u \in L^p(\mu)$, $f = k^{-1/p}u$ has $\int |f|^p\,d\mu_0 = \|u\|_p^p$ and $Kf = u$. And μ_0, μ have the same null sets. $gk^{1/q} \in L^q(\mu)$*:* for $1 < q < \infty$, $\int |gk^{1/q}|^q\,d\mu = \int |g|^q k\,d\mu = \int |g|^q\,d\mu_0 < \infty$. For $q = \infty$ ($p = 1$), $k^{1/q} = 1$ and $\|g\|_{L^\infty(\mu)} = \|g\|_{L^\infty(\mu_0)} < \infty$ because the null sets agree. □

Summarizing:

Theorem 7.2.3

For a σ-finite positive measure μ on a σ-algebra of X, $L^p(\mu)^* = L^q(\mu)$, where p, q are conjugate exponents and $1 \le p < \infty$.

Exercise 7.2.5

Let $X = \{a, b\}$ with $\mu(\{a\}) = 1$, $\mu(\{b\}) = \infty$. Is the dual of $L^1(\mu)$ equal to $L^\infty(\mu)$?

Solution [Supplement]. $f \in L^1(\mu)$ iff $|f(a)| + \infty \cdot |f(b)| < \infty$, i.e. $f(b) = 0$, so $L^1(\mu) \cong \mathbb{C}$ via $f \mapsto f(a)$ and $L^1(\mu)^* \cong \mathbb{C}$; but $\{b\}$ is not null, so $L^\infty(\mu) \cong \mathbb{C}^2$ (max norm). $g \mapsto \varphi_g$, $\varphi_g(f) = g(a)f(a)$, is onto $L^1(\mu)^*$ but not injective ($g = \chi_{\{b\}}$, $\|g\|_\infty = 1$, $\varphi_g = 0$): $L^1(\mu)^* \neq L^\infty(\mu)$, so σ-finiteness is needed in Theorem 7.2.3 for $p = 1$. □

Corollary 7.2.4

Let μ be a σ-finite positive measure on X and $1 \le p \le \infty$. If a measurable function g on X satisfies

$$h \in L^q(\mu) \implies \Big| \int_X gh \, d\mu \Big| \le M \|h\|_q \qquad (p, q \text{ conjugate}),$$

then $g \in L^p(\mu)$ and $\|g\|_p \le M$.

Proof. The hypothesis says $\varphi_g \in L^q(\mu)^*$ and $\|\varphi_g\| \le M$. So if $1 < p \le \infty$ (i.e. $1 \le q < \infty$), Theorem 7.2.3 applies and the isometry gives $\|g\|_p = \|\varphi_g\| \le M$. For $p = 1$, the hypothesis gives

$$\int_X |g| \, d\mu = \int_X g \cdot (\operatorname{sgn} g) \, d\mu \le M \|\operatorname{sgn} g\|_\infty \le M,$$

where $\operatorname{sgn} g = \overline{g}/|g|$ on $\{g \ne 0\}$ and 0 elsewhere. ■

Corollary 7.2.5

For every $f \in L^1(\mathbb{R})$ and $g \in L^p(\mathbb{R})$ $(1 \le p \le \infty)$, $f * g \in L^p(\mathbb{R})$ and $\|f * g\|_p \le \|f\|_1 \|g\|_p$.

Proof. For every $h \in L^q(\mathbb{R})$, writing $g_y(x) = g(x - y)$,

$$\begin{aligned} \Big| \int_{\mathbb{R}} (f * g)(x) h(x) \, dx \Big| &= \Big| \int_{\mathbb{R}} \int_{\mathbb{R}} f(y) g(x - y) \, dy \, h(x) \, dx \Big| \\ &\le \int_{\mathbb{R}} \Big| \int_{\mathbb{R}} g_y(x) h(x) \, dx \Big| \, |f(y)| \, dy \\ &\le \|g_y\|_p \|h\|_q \|f\|_1 = \|f\|_1 \|g\|_p \|h\|_q \end{aligned}$$

(Fubini, Hölder, translation invariance). Corollary 7.2.4 gives the result. ■

(To be careful, apply the computation first to $|f|, |g|, |h|$: Tonelli gives $\iint |f(y) g(x-y) h(x)| \, dy \, dx \le \|f\|_1 \|g\|_p \|h\|_q < \infty$, which justifies Fubini and shows $(f * g)(x)$ is defined a.e.)

7.3 Borel Measures on the Real Line

Motivation. Finite Borel measures on $\mathbb{R}$ come from increasing functions; the discrete, singular continuous and absolutely continuous parts of the measure match those of the function, the last recovered by differentiation.

7.3.1 Regular Borel measures and their distribution functions

We find all complex measures on the Borel sets $\mathcal{B}$ of $\mathbb{R}$ (§4.1), the **complex Borel measures**. As they are combinations of positive measures, consider positive Borel measures finite on bounded intervals: **regular positive Borel measures**.

For a regular positive Borel measure λ on the real line define $\alpha_\lambda : \mathbb{R} \to \mathbb{R}$ by

$$\alpha_\lambda(x) = \begin{cases} \lambda((0, x]), & x > 0, \\ 0, & x = 0, \\ -\lambda((x, 0]), & x < 0. \end{cases}$$

Then α_λ is clearly nondecreasing. It is also right-continuous. First let $x > 0$ and take any decreasing sequence $\langle h_n \rangle$ in $(0, x)$ converging to 0. Then

$$\alpha_\lambda(x + h_n) - \alpha_\lambda(x) = \lambda((0, x + h_n]) - \lambda((0, x]) = \lambda((x, x + h_n]),$$

and since $\bigcap_{n=1}^{\infty}(x, x+h_n] = \emptyset$ and $\lambda((x, x+h_1]) < \infty$, continuity from above gives

$$\lim_{n\to\infty}\{\alpha_\lambda(x+h_n) - \alpha_\lambda(x)\} = 0.$$

Thus $\lim_{h\to 0^+}\alpha_\lambda(x+h) = \alpha_\lambda(x)$. The cases $x = 0$ and $x < 0$ are handled the same way, so α_λ is right-continuous.

Exercise 7.3.1

For a regular positive Borel measure λ on the real line, show that $\alpha_\lambda : \mathbb{R} \to \mathbb{R}$ is a nondecreasing right-continuous function.

Solution [Supplement]**.** In all cases $\alpha_\lambda(b) - \alpha_\lambda(a) = \lambda((a, b]) \geq 0$ for $a < b$ (e.g. for $a < 0 < b$, $\lambda((0, b]) + \lambda((a, 0])$), so α_λ is nondecreasing. For $h_n \searrow 0$, $\alpha_\lambda(x+h_n) - \alpha_\lambda(x) = \lambda((x, x+h_n]) \to 0$ by continuity from above ($\lambda((x, x+h_1]) < \infty$); by monotonicity α_λ is right-continuous. □

7.3.2 Riemann–Stieltjes integrals

For bounded $f : [a, b] \to \mathbb{R}$ the Riemann–Stieltjes integral $\int_a^b f\, d\alpha_\lambda$ is built like the Riemann integral (Fact B.13): for a partition $P = \{x_0, \ldots, x_n\}$, with U_P, L_P as in Fact B.13, the sums are

$$\begin{aligned} U_a^b(f, P, \alpha_\lambda) &= \sum_{i=1}^{n} M_i(\alpha_\lambda(x_i) - \alpha_\lambda(x_{i-1})) = \sum_{i=1}^{n} M_i\lambda((x_{i-1}, x_i]) \\ &= \int_{[a,b]} U_P\, d\lambda - f(a)\lambda(\{a\}) = \int_{(a,b]} U_P\, d\lambda, \end{aligned}$$

and similarly $L_a^b(f, P, \alpha_\lambda) = \int_{(a,b]} L_P\, d\lambda$. Hence for every partition $P \in P[a, b]$

$$L_a^b(f, P, \alpha_\lambda) \leq \int_{(a,b]} f\, d\lambda \leq U_a^b(f, P, \alpha_\lambda).$$

If $f : \mathbb{R} \to \mathbb{R}$ is continuous with support in a bounded interval (A, B), then f is Riemann–Stieltjes integrable on $[A, B]$ with respect to α_λ, so

$$\int_{\mathbb{R}} f\, d\lambda = \int_A^B f(x)\, d\alpha_\lambda(x).$$

So on $C_c(\mathbb{R})$, integration against a regular positive Borel measure is Riemann–Stieltjes integration. The latter struggles with discontinuous integrands; the measure integrals instead struggle with improper integrals.

7.3.3 Continuity of α_λ

Applying Lebesgue–Radon–Nikodym to Borel measures on $\mathbb{R}$, we first ask when α_λ is continuous or absolutely continuous. As $\bigcap_n(x - h_n, x] = \{x\}$ for $h_n \searrow 0$,

$$\lambda(\{x\}) = \lim_{n\to\infty}\lambda((x - h_n, x]) = \lim_{n\to\infty}\{\alpha_\lambda(x) - \alpha_\lambda(x - h_n)\}.$$

So α_λ is left-continuous at x iff $\lambda(\{x\}) = 0$. A measure for which every point has measure 0 is a **continuous measure**. Since α_λ is always right-continuous:

Proposition 7.3.1

For a regular positive Borel measure λ on the real line, α_λ is continuous at x if and only if $\lambda(\{x\}) = 0$.

The next theorem explains the shared term "absolutely continuous". Both measures must live on one σ-algebra; $m|_{\mathcal{B}}$ is also written m.

Theorem 7.3.2

For a regular positive Borel measure λ on the real line, the following are equivalent:

(a) $\lambda \ll m$;

(b) α_λ is absolutely continuous on every bounded closed interval.

Proof. (a)⇒(b). If $\lambda \ll m$, by the Radon–Nikodym theorem there is a Borel function $h : \mathbb{R} \to [0, \infty]$ with

$$\lambda(E) = \int_E h(x)\,dx, \qquad E \in \mathcal{B}.$$

Fix a bounded closed interval I; we show α_λ is absolutely continuous on I. Since $\int_I h = \lambda(I) < \infty$, $h \in L^1(I, m)$. By Fact B.15, for every $\epsilon > 0$ there is $\delta > 0$ with

$$A \in \mathcal{B},\ m(A) < \delta \implies \lambda(A) < \epsilon.$$

This δ works in the definition of absolute continuity of functions. If $\{(a_i, b_i) : i = 1, \dots, n\}$ are disjoint intervals in I with $\sum_{i=1}^n (b_i - a_i) < \delta$, then $m(\bigcup_{i=1}^n (a_i, b_i]) < \delta$, so

$$\sum_{i=1}^n |\alpha_\lambda(b_i) - \alpha_\lambda(a_i)| = \sum_{i=1}^n \lambda((a_i, b_i]) = \lambda\Big(\bigcup_{i=1}^n (a_i, b_i]\Big) < \epsilon.$$

(b)⇒(a). Let α_λ be absolutely continuous on each bounded closed I, with δ_I for ϵ. If $m(A) = 0$ then $m(A \cap I) = 0$, so there are disjoint open intervals (c_n, d_n) with

$$A \cap I \subset \bigcup_{n=1}^\infty (c_n, d_n), \qquad \sum_{n=1}^\infty (d_n - c_n) < \delta_I.$$

Writing (a_n, b_n) for the intersection of (c_n, d_n) with the interior $\operatorname{int} I$ of I,

$$A \cap \operatorname{int} I \subset \bigcup_{n=1}^\infty (a_n, b_n) \subset I, \qquad \sum_{n=1}^\infty (b_n - a_n) < \delta_I.$$

Hence $\sum_{n=1}^N [\alpha_\lambda(b_n) - \alpha_\lambda(a_n)] < \epsilon$ for every N, so $\sum_{n=1}^\infty [\alpha_\lambda(b_n) - \alpha_\lambda(a_n)] \le \epsilon$. Since α_λ is continuous (being absolutely continuous), $\alpha_\lambda(b_n) - \alpha_\lambda(a_n) = \lambda((a_n, b_n))$ (no point masses), and

$$\lambda(A \cap \operatorname{int} I) \le \lambda\Big(\bigcup_{n=1}^\infty (a_n, b_n)\Big) = \sum_{n=1}^\infty \lambda((a_n, b_n)) = \sum_{n=1}^\infty [\alpha_\lambda(b_n) - \alpha_\lambda(a_n)] \le \epsilon.$$

So $\lambda(A \cap \operatorname{int} I) = 0$ for every bounded open interval $\operatorname{int} I$, and therefore $\lambda(A) = 0$. ∎

7.3.4 Regularity

We now look for the condition on α_λ for $\lambda \perp m$. First some properties of regular positive Borel measures on the real line.

Theorem 7.3.3 (regularity)

A regular positive Borel measure λ on the real line satisfies, for every Borel set $E \in \mathcal{B}$,

$$\lambda(E) = \inf\{\lambda(U) : U \supset E,\ U \text{ open}\},$$
$$\lambda(E) = \sup\{\lambda(K) : K \subset E,\ K \text{ compact}\}.$$

Proof. Open sets satisfy the first identity and, as increasing unions of compact sets, the second. Let $\mathcal{C}$ be the Borel sets satisfying both; show $\mathcal{C}$ is a σ-algebra.

Countable unions. Take $\langle E_n\rangle$ in $\mathcal{C}$ and $E = \bigcup_n E_n$. First the case $\lambda(E) < \infty$: choose open $U_n \supset E_n$ with $\lambda(U_n) < \lambda(E_n) + \epsilon/2^n$ and let U be their union. Since $U \setminus E \subset \bigcup_n (U_n \setminus E_n)$,

$$\lambda(U) - \lambda(E) = \lambda(U \setminus E) \le \sum_{n=1}^{\infty} \lambda(U_n \setminus E_n) < \epsilon,$$

so E satisfies the first identity. If $\lambda(E) = \infty$ the first identity is trivial. For the second, for each j choose compact sets $\langle K_{ij} : i = 1, 2, \dots\rangle$ with

$$K_{ij} \subset E_j, \qquad \lim_{i\to\infty} \lambda(K_{ij}) = \lambda(E_j).$$

With $K_n = \bigcup_{i,j=1}^{n} K_{ij}$ we can show $\lambda(K_n) \to \lambda(E)$, so $E \in \mathcal{C}$ (Exercise 7.3.2).

Complements. For $E \in \mathcal{C}$ take open $U_n \supset E$ decreasing with $\lambda(U_n) \to \lambda(E)$; the compact $L_n = [-n, n] \setminus U_n \subset E^c$ have $\lambda(L_n) \to \lambda(E^c)$ (Exercise 7.3.2). For the first identity for E^c we first show

$$I \text{ a bounded closed interval},\ I \cap E \in \mathcal{C} \implies I \setminus E \in \mathcal{C}.$$

Choose an open $U \supset I \cap E$ with $\lambda(U) < \lambda(I \cap E) + \epsilon$. Then $K = I \setminus U$ is compact, $K \subset I \setminus E$, and since $\lambda(I) < \infty$,

$$\lambda(K) = \lambda(I) - \lambda(I \cap U) \ge \lambda(I) - \lambda(U) > \lambda(I) - \lambda(I \cap E) - \epsilon = \lambda(I \setminus E) - \epsilon,$$

so $I \setminus E$ satisfies the second identity. For the first, choose a bounded open interval $J \supset I$ with $\lambda(J) < \lambda(I) + \epsilon$, and a compact $K \subset I \cap E$ with $\lambda(K) > \lambda(I \cap E) - \epsilon$. Put $U = J \setminus K$; then U is open, $U \supset I \setminus E$, and

$$\lambda(U) = \lambda(J) - \lambda(K) < \lambda(I) + \epsilon - \lambda(I \cap E) + \epsilon = \lambda(I \setminus E) + 2\epsilon,$$

so $I \setminus E$ satisfies the first identity.

If $E \in \mathcal{C}$, then $E \cap [-n, n] \in \mathcal{C}$ for each n (one checks this as above), so $[-n, n] \setminus E \in \mathcal{C}$. Since $\mathcal{C}$ is already closed under countable unions, $E^c = \bigcup_n ([-n, n] \setminus E) \in \mathcal{C}$. ∎

Exercise 7.3.2

Show that every open set is an increasing union of compact sets, and in the proof of Theorem 7.3.3 that $\lambda(K_n) \to \lambda(E)$ and $\lambda(L_n) \to \lambda(E^c)$.

Solution [Supplement]. *Open sets.* For open $U \subset \mathbb{R}$ ($U \neq \mathbb{R}$) let $C_n = \{x \in U : |x| \le n,\ d(x, U^c) \ge \frac{1}{n}\}$; for $U = \mathbb{R}$ let $C_n = [-n, n]$. Each C_n is closed and bounded, hence compact, $C_n \subset C_{n+1}$, and every $x \in U$ has $d(x, U^c) > 0$, so $x \in C_n$ for large n.

$\lambda(K_n) \to \lambda(E)$. $K_n = \bigcup_{i,j\le n} K_{ij}$ is compact, $K_n \subset K_{n+1}$ and $K_n \subset E$, so $L = \lim_n \lambda(K_n) \le \lambda(E)$ exists. If some $\lambda(E_j) = \infty$, then $L \ge \lambda(K_{ij}) \to \infty$ as $i \to \infty$, so $L = \lambda(E) = \infty$. Otherwise fix N and i. For $n \ge \max(i, N)$, $K_n \supset \bigcup_{j\le N} K_{ij}$, and

$$\lambda\Big(\bigcup_{j\le N} E_j\Big) - \lambda\Big(\bigcup_{j\le N} K_{ij}\Big) \le \sum_{j\le N} \lambda(E_j \setminus K_{ij}) = \sum_{j\le N} \big(\lambda(E_j) - \lambda(K_{ij})\big) \xrightarrow[i\to\infty]{} 0.$$

Hence $L \geq \lambda(E_1 \cup \cdots \cup E_N)$ for every N, and continuity from below gives $L \geq \lambda(E)$.

$\lambda(L_n) \to \lambda(E^c)$. Suppose $\lambda(E) < \infty$. $L_n = [-n, n] \setminus U_n$ is compact, increasing (as U_n decreases), and $L_n \subset E^c$. Since $([-n, n] \cap E^c) \setminus L_n = [-n, n] \cap U_n \cap E^c \subset U_n \setminus E$,

$$0 \leq \lambda([-n, n] \cap E^c) - \lambda(L_n) \leq \lambda(U_n \setminus E) = \lambda(U_n) - \lambda(E) \to 0,$$

while $\lambda([-n, n] \cap E^c) \to \lambda(E^c)$ (continuity from below), so $\lambda(L_n) \to \lambda(E^c)$. (If $\lambda(E) = \infty$, apply this to $E \cap [-N, N]$, as at the end of the proof.) □

Corollary 7.3.4

If $\lambda \ll m$ and $\alpha'_\lambda = 0$ almost everywhere, then $\lambda = 0$.

Proof. By Theorem 7.3.2, α_λ is absolutely continuous, so Fact B.16 gives

$$\lambda((a, b)) = \alpha_\lambda(b) - \alpha_\lambda(a) = \int_a^b \alpha'_\lambda(x)\, dx = 0.$$

Hence $\lambda(U) = 0$ for every open U (a countable disjoint union of open intervals), and Theorem 7.3.3 gives $\lambda = 0$. ■

7.3.5 Singular measures

Theorem 7.3.5

For a regular positive Borel measure λ on the real line, the following are equivalent:

(a) $\lambda \perp m$;

(b) $\alpha'_\lambda = 0$ almost everywhere.

Proof. (b)⇒(a). Assume $\alpha'_\lambda = 0$ a.e. Write the Lebesgue decomposition $\lambda = \lambda_a + \lambda_s$; then $\alpha_\lambda = \alpha_{\lambda_a} + \alpha_{\lambda_s}$, both nondecreasing, so $0 \leq \alpha'_{\lambda_a} \leq \alpha'_\lambda = 0$ and hence $\alpha'_{\lambda_a} = 0$ almost everywhere. Since $\lambda_a \ll m$, Corollary 7.3.4 gives $\lambda_a = 0$, so $\lambda = \lambda_s$ is singular.

(a)⇒(b). Assume $\lambda \perp m$ and put

$$\nu(E) = \int_E \alpha'_\lambda\, dm, \qquad E \in \mathcal{B}. \tag{15}$$

Then obviously $\nu \ll m$. On the other hand, by Fact B.14,

$$\nu((a, b]) = \int_{(a,b]} \alpha'_\lambda\, dm \leq \alpha_\lambda(b) - \alpha_\lambda(a) = \lambda((a, b]).$$

Open sets are countable disjoint unions of intervals $(a, b]$, so $\nu(U) \leq \lambda(U)$ for open U, and Theorem 7.3.3 gives $\nu \leq \lambda$ on Borel sets, so $\nu \ll \lambda$. With $\lambda \perp m$, $\nu \perp m$ (Exercise 7.3.3), and $\nu \ll m$ forces $\nu = 0$. As $\alpha'_\lambda \geq 0$ a.e., $\alpha'_\lambda = 0$ a.e. ■

Exercise 7.3.3

Show that if $\nu \ll \lambda$ and $\lambda \perp m$, then $\nu \perp m$.

Solution [Supplement]. Choose disjoint A, B with $\lambda(E) = \lambda(E \cap A)$ and $m(E) = m(E \cap B)$ for all E. Then $\lambda(E \setminus A) = 0$, so $\nu(E \setminus A) = 0$ by $\nu \ll \lambda$, i.e. $\nu(E) = \nu(E \cap A)$. Together with $m(E) = m(E \cap B)$, the disjoint pair A, B shows $\nu \perp m$. □

7.3.6 The Radon–Nikodym derivative is α'_λ

For an arbitrary regular positive Borel measure λ with Lebesgue decomposition $\lambda = \lambda_a + \lambda_s$, define ν by (15). By Theorem 7.3.5,

$$\alpha'_\lambda = \alpha'_{\lambda_a} + \alpha'_{\lambda_s} = \alpha'_{\lambda_a}$$

almost everywhere, and by Fact B.16 (α_{λ_a} absolutely continuous by Theorem 7.3.2)

$$\nu((c,d)) = \int_{(c,d)} \alpha'_\lambda \, dm = \int_{(c,d)} \alpha'_{\lambda_a} \, dm = \alpha_{\lambda_a}(d) - \alpha_{\lambda_a}(c) = \lambda_a((c,d)).$$

Hence $\nu(U) = \lambda_a(U)$ for every open U, and since ν is also regular, $\nu = \lambda_a$ (Theorem 7.3.3). That is,

$$\lambda_a(E) = \int_E \alpha'_\lambda \, dm$$

for every Borel set $E \in \mathcal{B}$: $\alpha'_{\lambda_a} = \alpha'_\lambda$ is precisely the Radon–Nikodym derivative of λ_a with respect to m.

Corollary 7.3.6

If a regular positive Borel measure λ on the real line is absolutely continuous with respect to Lebesgue measure m, then $\left[\frac{d\lambda}{dm}\right] = \alpha'_\lambda$ almost everywhere.

7.3.7 Discrete, continuous singular, and absolutely continuous parts

For a regular positive Borel measure λ put

$$D = \{x \in \mathbb{R} : \lambda(\{x\}) > 0\},$$

and for $E \in \mathcal{B}$

$$\lambda_d(E) = \lambda(E \cap D), \qquad \lambda_c(E) = \lambda(E \setminus D).$$

For each n, $\lambda(D \cap [-n,n]) \le \lambda([-n,n]) < \infty$, so $D \cap [-n,n]$ is countable (for each k, at most $k\lambda([-n,n])$ points have mass $> 1/k$); hence D is countable, $D = \{x_n : n = 1,2,\dots\}$. Then for every $E \in \mathcal{B}$

$$\lambda_d(E) = \sum_{x_n \in E} \lambda(\{x_n\}) = \sum_{n=1}^{\infty} \lambda(\{x_n\}) \delta_{x_n}(E),$$

so λ_d is a sum of Dirac measures, and α_{λ_d} jumps at each $x \in D$; α_{λ_c} is continuous (Proposition 7.3.1), and if $\lambda_c \perp m$ its derivative is 0 a.e. (Theorem 7.3.5), as for the Cantor function.

Summary

Every regular positive Borel measure λ on the real line is the sum of three measures:

$$\lambda = \lambda_a + \lambda_s + \lambda_d,$$

an absolutely continuous λ_a, a continuous singular λ_s and a discrete λ_d, whose functions are absolutely continuous (α_a), continuous with derivative 0 a.e. (α_s), and a jump function (α_d). Then $\alpha = \alpha_a + \alpha_s + \alpha_d$ and

$$\lambda_a(E) = \int_E \alpha'_a \, dm = \int_E \alpha' \, dm, \qquad E \in \mathcal{B}.$$

Given a regular positive Borel measure λ, there is a corresponding nondecreasing right-continuous function α, and the two linear functionals on $C_c(\mathbb{R})$

$$f \mapsto \int_{\mathbb{R}} f \, d\lambda, \qquad f \mapsto \int_{\operatorname{supp} f} f(x) \, d\alpha(x)$$

coincide (Borel measure integral and Riemann–Stieltjes integral). Conversely, every nondecreasing function gives a regular positive Borel measure (next chapter).